

Low-Cost Bridge Vibration Anomaly Detection System Robust to Temperature and Humidity Changes

1 Topic

1.1 Background and Need

Sudden collapses of aging bridges — a recurring social problem

In April 2023, the walkway of the roughly 30-year-old Jeongja Bridge in Bundang suddenly collapsed while citizens were walking across it, killing one person and injuring another. The collapse of a bridge that looked intact from the outside was linked to corrosion and damage that had accumulated over a long period in its cantilever members. Warning signs such as cracking and sagging had been observed in inspections before the accident but did not lead to sufficient repair or reinforcement. This case shows that the mere existence of periodic inspection makes it difficult to track changes in structural condition continuously. Sudden collapses of aging bridges in service are not unique to Jeongja Bridge. The collapse of the Fern Hollow Bridge in Pittsburgh, USA, in January 2022 is a case in which progressive corrosion had already been flagged in periodic inspections, but repairs were delayed and the bridge finally fell while in use during the morning commute, revealing that having passed periodic inspections alone does not prevent accidents.



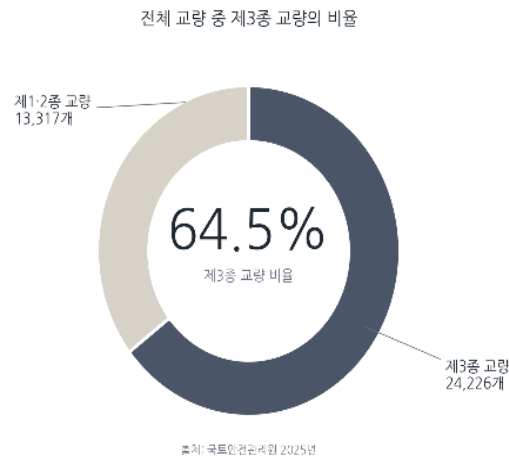
[Fig. 1: Jeongja Bridge collapse site]



[Fig. 2: Fern Hollow Bridge collapse]

The two cases show that internal damage such as corrosion, cracking and fatigue can progress over long periods in aging bridges in service. Because such damage can lead to changes in dynamic properties such as stiffness, damping and natural frequency, it may be observable as an anomaly through vibration-based monitoring. However, a structure's vibration characteristics change not only with actual damage but also with environmental variables such as temperature and humidity, so even when a change in vibration is observed it is hard to tell whether it comes from damage or from environmental change. The problem therefore remains that periodic inspection alone can hardly capture, continuously between inspections, dynamic changes that do not show on the surface, and that separating environmental effects from observed vibration changes to pick out anomalies is also not easy.

According to the Korea Authority of Land & Infrastructure Safety's 2025 statistics by facility class, Korea has 37,543 bridges — 5,392 Class 1, 7,925 Class 2 and 24,226 Class 3 — of which Class 3 bridges account for about 64.5%. This shows that the scope of safety management extends not only to large bridges but also to relatively small ones. The need for low-cost continuous screening technology applicable to small and medium bridges is therefore growing.



[Fig. 3: Share of Class 3 bridges among all bridges in Korea]

The continuous-monitoring blind spot of small and medium bridges

As the need for safety management of aging and vulnerable facilities grew, the scope of precision safety diagnosis was also expanded from December 4, 2025. Class 2 facilities rated D or E in a precision safety inspection, and Class 2 and 3 facilities rated C, D or E in a safety inspection that had not received a precision safety diagnosis since 30 years after completion, were added to its scope. Precision safety diagnosis, however, remains a costly and labor-intensive procedure performed only on a fixed cycle — every 4 to 6 years depending on the safety grade — so a time gap opens between inspections during which the condition is not checked. If dynamic changes in a structure accumulate slowly during this gap while it passes visual inspection, they are hard to detect before the next inspection. As a result, pedestrian overpasses, small footbridges and aging small and medium bridges lack a low-cost supplementary means of continuously detecting changes in condition between inspections.

종류 안전등급	A 등급	B·C 등급	D·E 등급
정기안전점검	반기에 1회 이상	반기에 1회 이상	1년에 3회 이상
정밀안전점검 (건축물)	4년에 1회 이상	3년에 1회 이상	2년에 1회 이상
긴급안전점검	재난 또는 재해 발생 우려 있는 경우	재난 또는 재해 발생 우려 있는 경우	재난 또는 재해 발생 우려 있는 경우
정밀안전진단	6년에 1회 이상	5년에 1회 이상	4년에 1회 이상

[Fig. 4: Inspection cycles and targets by facility grade]

No low-cost screening technology before precision diagnosis

Existing SHM (Structural Health Monitoring) systems often rely on precision measurement equipment such as ultra-precise piezoelectric accelerometers, high-performance data-acquisition hardware, cloud-based analysis infrastructure or distributed fiber-optic sensors. For example, the smart fiber-optic BOCDA sensor of the Korea Research Institute of Standards and Science (KRISS) has the advantage of precisely measuring strain and load position in a structure. Because such precision sensor systems require fiber-optic measurement infrastructure and dedicated equipment, however, their cost and installation constraints make them hard to apply widely and cheaply to small and medium aging bridges already in service. As a result, low-cost screening technology that can continuously monitor changes in the condition of small and medium bridges before the precision-diagnosis stage is effectively missing. Bridge safety management therefore needs a multi-stage concept in which continuous screening (first stage) filters out anomaly candidates before precision safety inspection and diagnosis. If low-cost screening fills this first stage, limited inspection staff and budgets can be concentrated on suspect bridges where anomaly candidates have been observed.

1.2 Topic and Purpose

This work aims to address the three problems summarized above: that periodic inspection alone can hardly capture, continuously, dynamic changes that do not show on the surface; that there is no means of continuously detecting changes in condition that accumulate between inspection cycles; and that existing precision measurement systems are hard to apply widely and cheaply to small and medium aging bridges. This work, however, is not a system that predicts collapse directly or replaces precision safety diagnosis. It is a low-cost first-stage (supplementary) screening system that observes vibration characteristics together with environmental variables using a low-cost MEMS accelerometer and a temperature/humidity sensor, compensates for normal variation explainable by temperature and humidity, detects the remaining anomaly candidates early, and indicates which structures should be prioritized for precision inspection. In other words, it does not replace precision diagnosis but is positioned as a supplementary means of filling the time gap between periodic inspection and precision diagnosis. The purpose of this work is to develop a low-cost, machine-learning-based bridge vibration screening system that addresses both the continuous-monitoring blind spot of small and medium bridges and the absence of low-cost screening technology. The specific goals are three. First, collect vibration data using an ADXL355Z-based MEMS accelerometer and a Raspberry Pi 3 Model A+. Second, record temperature and relative humidity with a DHT22 sensor and learn the relationship between temperature/humidity and natural frequency from normal-state data to reduce false alarms caused by environmental change. Third, implement an explainable alarm system that, rather than simply outputting whether there is an anomaly, also shows in which frequency band (or band around a mode) the normal baseline was departed from.

The reasons for the three goals are as follows. First, it is practically difficult to obtain training data containing both normal and anomalous states, because damaged-state data cannot be created deliberately on a bridge in service; this work therefore adopts unsupervised learning trained only on normal data. Next, temperature/humidity compensation is needed because a bridge's natural frequency rises and falls naturally with air temperature and humidity even without damage [2]. Mistaking this natural variation for damage increases false alarms, so the part explained by temperature and humidity must be removed first and only the remaining change examined. Finally, the alarm must be explainable because a single "anomaly" signal makes it hard for field inspectors to judge where to look more closely. The system therefore also shows in which band the normal baseline was departed from, helping to set the direction of the follow-up precision inspection.

Approach

The core technology of this work is unsupervised anomaly detection on vibration data. When the acceleration sensor measures vibration, a machine-learning model first learns the vibration patterns of the normal state and then judges vibration that departs strongly from those patterns to be an anomaly candidate. Because simple anomaly detection alone leaves the problem of false alarms caused by environmental change, the following three functions are combined.

1. Low-cost vibration measurement: the x-, y- and z-axis acceleration of the model bridge is recorded with an ADXL355Z-based 3-axis MEMS accelerometer module and a Raspberry Pi 3 Model A+. Because this work uses a foam-board model, the emphasis is on fixing the sensor firmly against the structure; for application to a real steel bridge, a magnetic bracket can also be used.
2. Environmental compensation: normal variation due to the environment is first compensated using the relationship between temperature/humidity and vibration characteristics learned from normal data, and only the residuals not explained by temperature and humidity are used for anomaly detection. This reduces false alarms caused by environmental change.
3. Explainable alarms: when an anomaly is detected, the system does not output only an overall anomaly score but visualizes in which frequency band (or band around a mode) a large reconstruction error occurred. The system does not definitively diagnose the cause of damage; it provides interpretive information so that field inspectors can judge the direction of the follow-up precision inspection. The concrete signal-processing, regression-compensation and anomaly-detection procedure that links these three functions from data collection to output is covered step by step in 2.3 Operating Principle.

1.3 Similar Products (Research) and Differentiation

SHM systems based on low-cost MEMS accelerometers

An SHM study combined a Raspberry Pi with a low-cost MEMS accelerometer (ADXL355) and showed that a structure's natural frequencies, mode shapes and damping can be identified without ultra-precise piezoelectric sensors costing tens of millions of won [1]. That study, however, focuses on identifying dynamic

characteristics and does not separately address how environmental variables such as temperature and humidity, or changes in operational loading such as vehicle traffic, affect anomaly detection. This work introduces an environment-adaptive pipeline that learns the relationship between temperature/relative humidity and natural frequency by multivariate regression and then performs residual-based anomaly detection, reducing false alarms caused by temperature/humidity change.

Distributed fiber-optic sensor systems

The BOCD (Brillouin Optical Correlation Domain Analysis) fiber-optic sensor system developed by the Korea Research Institute of Standards and Science (KRISS) uses optical fiber to precisely measure strain and load position in a structure [4]. It has the advantage of identifying deformation or load concentration at specific points with high spatial resolution. This work, by contrast, aims not at fiber-based precision strain measurement but at continuously observing changes in a structure's dynamic vibration characteristics with a low-cost MEMS accelerometer. The two technologies therefore differ in the physical quantity measured and their purpose, and this work focuses on low-cost first-stage alerts and on indicating precision-inspection priorities for small and medium bridges.

Unsupervised-learning SHM algorithms

The Swiss Z24 Bridge Benchmark provided by KU Leuven is one of the most widely used public datasets in vibration-based SHM and includes long-term monitoring data and progressive damage-test data [3]. Recently, many unsupervised models — Autoencoder, One-Class SVM, Isolation Forest, VAE and others — have been compared on the Z24 dataset [5][6]. Most of these studies, however, stop at outputting an anomaly score or whether a threshold is exceeded. This work decomposes the Autoencoder's reconstruction error by input feature or frequency band to visualize in which band the normal baseline was strongly departed from, and compares anomaly scores before and after temperature/humidity compensation to distinguish normal variation explainable by environmental variables from unexplained structural anomaly candidates.

Environmental-compensation research on long-term data from a full-scale bridge

The openLAB Research Bridge public dataset, from the long-term monitoring of a full-scale research bridge in Bautzen, Germany, includes environmental measurements such as air temperature and humidity along with acceleration responses, making it well suited to analyzing how seasonal change and daily temperature ranges affect a structure's vibration characteristics (dataset details in 2.7) [8]. The contribution of this work, however, is not the result of analyzing public data but the linking of low-cost sensor-based data collection, temperature/humidity compensation, anomaly judgment and band-wise explanation into one pipeline. The experiment on our own model bridge collects real data with an ADXL355Z, a DHT22 and a Raspberry Pi 3 Model A+ and confirms changes in vibration characteristics with changes in mass distribution. openLAB is used as supplementary validation data to check whether this environmental-compensation principle also holds on long-term monitoring data from a full-scale bridge. This study is therefore distinctive in that it is not a reanalysis of public data alone but a screening system that combines a low-cost hardware implementation with validation on public data from a full-scale bridge.

These distinctions can be summarized along three axes: direct low-cost hardware implementation, temperature/humidity compensation, and band-wise explainability. Unlike low-cost MEMS studies, this work reduces false alarms through temperature/humidity compensation; unlike fiber-optic measurement, it performs continuous screening with inexpensive general-purpose parts; and unlike unsupervised studies that output only an anomaly score, it also reports the contribution of each band to the reconstruction error.

2 Building the System

2.1 Objective

This system is a supplementary vibration-screening system for early detection of vibration-anomaly candidates in aging small and medium bridges, based on low-cost, low-power hardware and unsupervised learning. It aims to collect vibration, temperature and relative humidity

with an ADXL355Z-based MEMS accelerometer and a DHT22 temperature/humidity sensor, and to reduce environment-driven false alarms using the temperature/humidity-frequency relationship learned in the normal state. It then detects anomaly candidates that depart from the normal vibration pattern with Autoencoder-based unsupervised learning and, when an anomaly occurs, shows which frequency bands (or bands around modes) have large reconstruction errors, helping to prioritize precision inspection.

2.2 Design

Hardware

The system is built as a hardware acquisition unit and a software analysis unit. The hardware acquisition unit consists of an ADXL355Z-based 3-axis MEMS accelerometer module, a DHT22 temperature/humidity sensor [7], a Raspberry Pi 3 Model A+, a power supply and a foam-board bridge model. Because the Raspberry Pi can store data on its own microSD card, a separate microSD storage module is optional rather than required.

Component	Role	Design rationale
ADXL355Z 3-axis accelerometer	x/y/z-axis vibration measurement	Detects changes in vibration characteristics and natural frequency
DHT22 temperature/humidity sensor	Temperature and relative humidity	Compensates normal vibration variation due to temp/humidity
Raspberry Pi 3 Model A+	Data collection, storage, first-pass processing	Runs Python data logging and analysis code
Power supply	Powers the system	Stable power for long experiments
Foam-board bridge model	Model structure for vibration and mass-distribution tests	Represents a real bridge; structural-condition tests such as changing the mass distribution of weights on the foam-board deck

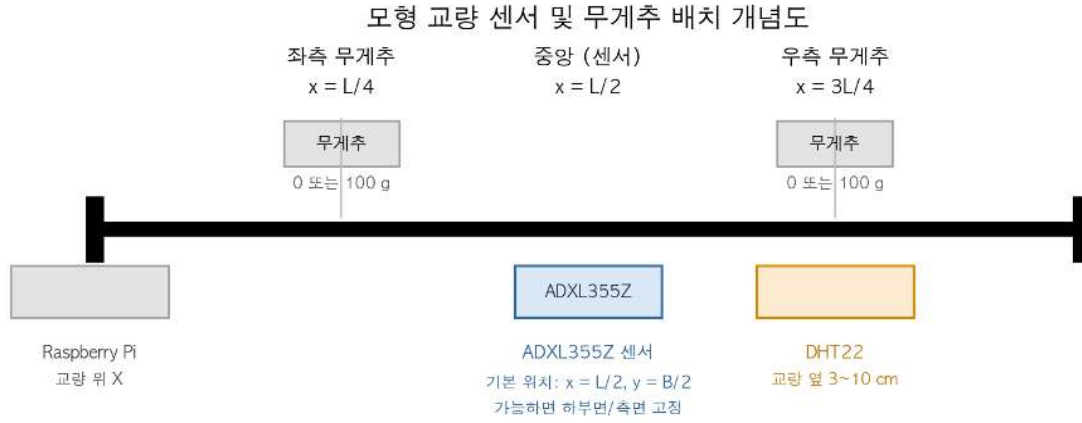
Software

The software analysis unit consists of an acquisition program and an analysis program. The acquisition program runs in Python on the Raspberry Pi 3 Model A+, receiving 3-axis acceleration from the ADXL355Z-based MEMS accelerometer at 200 Hz and temperature and relative humidity from the DHT22 at about 2-second intervals, and writes them to a single CSV file. The analysis program takes this file, splits the signal into fixed time windows, performs FFT and feature extraction, and then proceeds through residual calculation by multivariate regression on temperature T and relative humidity RH, standardization, reconstruction-error-based anomaly scoring, a 99th-percentile threshold decision, and a band-contribution plot. Because the two programs are linked by the same CSV format, the flow runs from collection to decision and explanation without any separate conversion step.

Sensor placement

The ADXL355Z accelerometer is attached at the lengthwise center of the model bridge. With the full span of the model L and its width B, the default sensor position is $x = L/2$, $y = B/2$. In a simply supported model bridge the first bending response is largest at mid-span, so when only a single accelerometer is used, mid-span is the best place to observe changes in the dynamic properties of the whole structure. The sensor is fixed with its x axis along the bridge, its y axis across it and its z axis vertical; vibration features are computed for all three axes, but the z axis, which best reflects vertical bending, is the main axis of analysis.

Because the DHT22 is not a vibration sensor, it is not placed on the model bridge itself but in the same test environment within 3-10 cm of the ADXL355Z. It is mounted on a small fixed stand beside the bridge to record temperature and relative humidity without affecting the model's mass or stiffness. The Raspberry Pi 3 Model A+ is likewise not placed on the bridge but on a table or fixed stand beside it. Sensor cables are fixed near the supports with a little slack around the sensor so that cable tension does not affect the vibration response.



[Figure 1: Layout of sensors and weights on the model bridge]

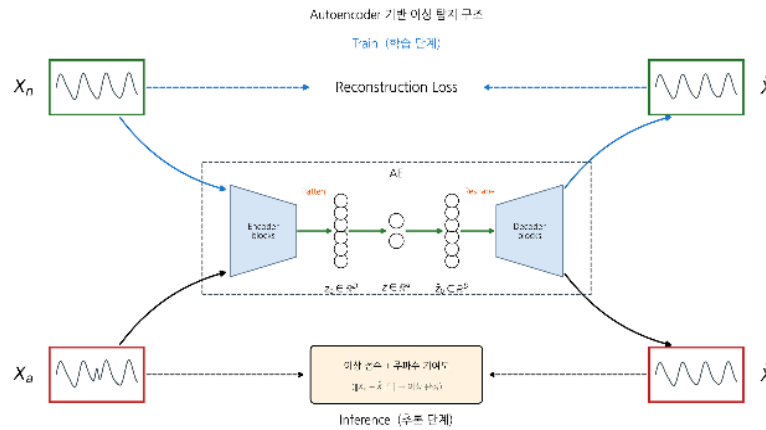
2.3 Operating Principle

1. Vibration and temperature/humidity acquisition: acceleration data are collected from the ADXL355Z-based 3-axis MEMS accelerometer, and temperature T and relative humidity RH are recorded from the DHT22. In the model-bridge experiment, the actual natural-frequency range is first checked in a preliminary test by tapping the model, and if necessary the analysis band is extended up to 100 Hz, the Nyquist limit of the 200 Hz sampling rate.

2. Signal preprocessing and frequency-band feature extraction: the DC offset is removed from the time-domain acceleration signal, band-pass filtering is applied where necessary, and the frequency spectrum is computed with FFT or STFT. For each analysis segment and axis, RMS, standard deviation, peak-to-peak, spectral entropy, peak frequency and peak power, and band energies (default 0-3, 3-10, 10-20 Hz) are computed, and temperature T and relative humidity RH are used as model inputs or for environmental compensation. In a simply supported model bridge the z-axis peak frequency indicates the first natural frequency, so the first natural frequency is estimated from this peak. On the model bridge, the band boundaries can be adjusted to the peak frequency observed in the normal-state preliminary test. The band-wise analysis is used not to locate damage directly but as an auxiliary indicator explaining which vibration characteristic departed from the normal baseline.

3. Multivariate regression compensation of the normal natural frequency by temperature and humidity: on normal-state data, a multivariate regression model is trained with temperature T and relative humidity RH as explanatory variables to predict the first natural frequency of the healthy structure. At inference, the current temperature and humidity are fed into the regression model to compute the normal-state prediction $\hat{f} = g(T, RH)$, and the difference from the measurement, $r = f - \hat{f}$, is defined as the residual corrected for the temperature/humidity effect. In this work, the same (T, RH) regression and residual calculation are applied not only to the first natural frequency but to each of the vibration features extracted above, and the resulting residual features together form the input to the anomaly-detection model. Because these residuals can include not only structural anomaly candidates but also sensor noise, changes in mounting and operational loading, they are used in this study as a criterion for detecting anomaly candidates rather than as a confirmation of damage.

4. Unsupervised anomaly detection with an Autoencoder: the Autoencoder is trained only on residual features from the normal state. Normal data are reconstructed well, while data that depart from the normal pattern have a large reconstruction error. The anomaly threshold is set at an upper percentile (default 99th) of the reconstruction-error distribution of normal validation data. Each residual feature is standardized before training so that features with different units are reflected evenly. Because the model compresses the input into a low-dimensional latent representation and reconstructs it, it learns only the correlation structure of normal data. Since both training and thresholding use only normal data, continuous screening can begin without obtaining any damage data.



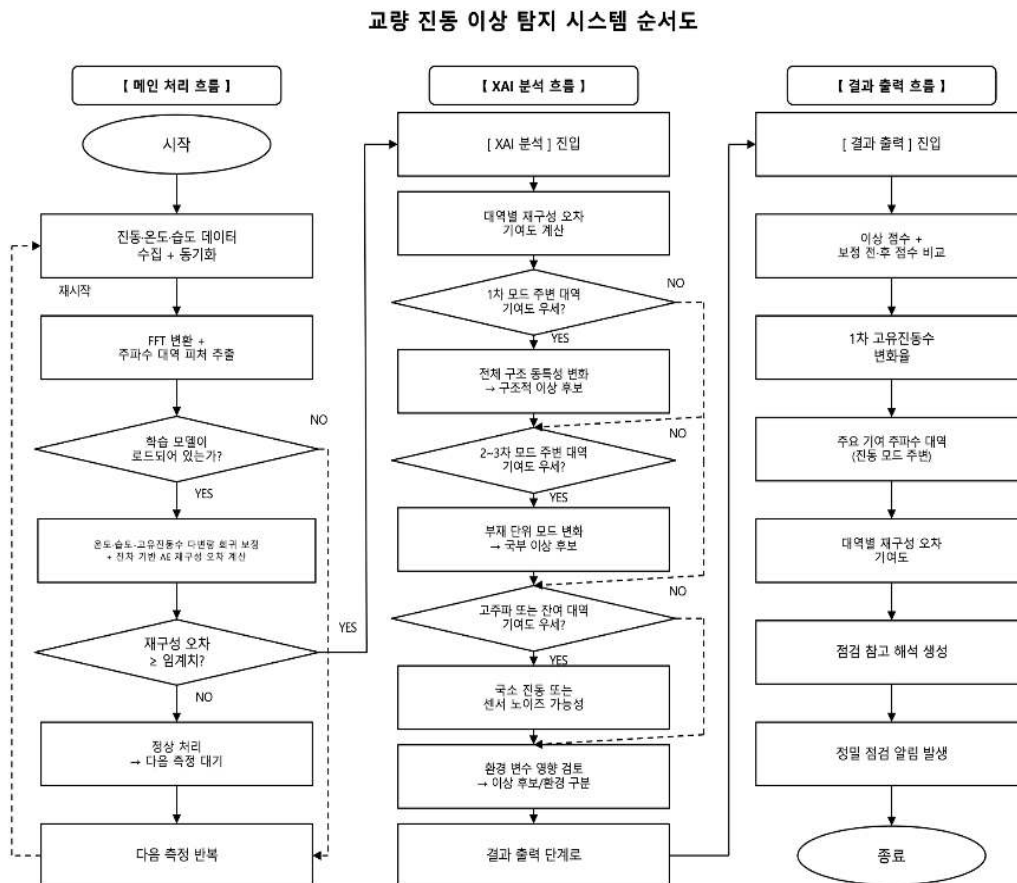
[Figure 2: Autoencoder-based anomaly detection]

5. Band-wise reconstruction-error contribution: when an anomaly is detected, the system does not output only an overall anomaly score but also analyzes in which frequency band (or band around a mode) a large reconstruction error occurred. A band's contribution is computed as “that band's reconstruction error ÷ total reconstruction error.” SHAP is kept as an optional feature that can additionally be applied to a separate auxiliary model, and the default method of explanation is the band-wise reconstruction-error contribution.

6. Output and precision-inspection alerts: when the anomaly score exceeds the threshold, an alarm is raised. The alert outputs the anomaly score, a comparison of anomaly scores before and after temperature/humidity compensation, the main contributing frequency band or band around a mode, the rate of change in the first natural frequency, the band-wise reconstruction-error contributions, and an interpretation for inspectors.

2.4 System Flowchart

The operating structure that links these modules as software is shown in the system flowchart below.



[Figure 3: Flowchart of the bridge vibration anomaly detection system]

2.5 Development Process

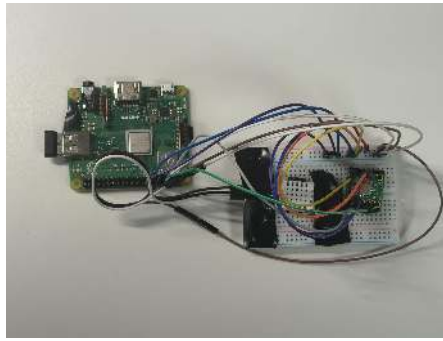
Step	Work	Output
Step 1 (06/03)	Connected ADXL355Z, DHT22 and Raspberry Pi 3 Model A+ and checked operation	Sensor connection test results
Step 2 (06/03)	Built the model bridge; decided sensor mounting positions	Model bridge, sensor mounting photos
Step 3 (06/03)	Collected 0-0 normal baseline data (mass-condition notation in 3.1)	Normal raw CSV
Step 4 (06/04)	Collected data for mass-distribution conditions 0-0, 1-0, 1-1	Raw CSV for each of 3 conditions
Step 5 (06/04)	Collected data by environmental condition (outdoor temperature excluded for reliability, see 3.2)	CSV for baseline, simultaneous temp/humidity change, humidity-centered change
Step 6 (06/04)	Organized data for 9 conditions	3 mass conditions × 3 environmental conditions Feature dataset
Step 7 (06/04)	FFT/STFT and feature extraction	Frequency feature dataset
Step 8 (06/05)	Autoencoder training and anomaly scoring	Anomaly scores, threshold, detection results
Step 9 (06/05)	Band-wise reconstruction-error contribution analysis	XAI contribution plots
Step 10 (06/05~06/12)	Results summarized; report written	Final report, presentation

2.6 Components Built and Key Implementation

Hardware build

The hardware and model bridge actually built and assembled are shown below.

 ADXL355Z 3-axis MEMS accelerometer	 DHT22 temperature/humidity sensor	 Raspberry Pi 3 Model A+
 Foam-board bridge model (about 55 cm long)		



Finished Raspberry Pi module

*The full source code, including the core source code, is attached as a separate archive (zip).

2.7 openLAB Analysis Results

This section presents the results of applying the proposed anomaly-detection pipeline to the openLAB research-bridge monitoring dataset [8] published in TU Dresden's research data repository (OPARA). openLAB is a full-scale research bridge built in Bautzen, Saxony, Germany: a three-span prestressed concrete bridge about 45 m long. The dataset was collected for about nine months, from February to October 2024, in the undamaged reference state, and provides event-triggered acceleration together with air temperature, humidity and solar radiation, so environmental variation from seasonal change and daily temperature ranges accumulates in it. Because the whole period is undamaged, every alarm exceeding the threshold can be interpreted as a false alarm, which makes the dataset well suited to quantifying the change in false-alarm rate due to environmental compensation. In the analysis, vibration features were extracted from the acceleration records as in 2.3 and linked to the air-temperature and humidity data by time. The normal samples were split in time order into 263 for training and 263 for evaluation, the anomaly threshold was set at the 99th percentile of the training reconstruction error, and three models — uncompensated AE, temperature-compensated AE and temperature/humidity-compensated AE — were compared under identical conditions. The results for each model are shown in the table below.

Model	Training samples	Evaluation samples	Anomaly rate, training	False-alarm rate, evaluation
Uncompensated AE	263	263	0.011	0.019
Temperature-compensated AE	263	263	0.011	0.015
Temp/humidity-compensated AE	263	263	0.011	0.015

All three models had a training anomaly rate of 0.011 (about 1.1%), consistent with setting the threshold at the 99th percentile of the training reconstruction error. Because the entire evaluation period is in the undamaged reference state, every threshold exceedance is a false alarm. The false-alarm rate was 1.90% for the uncompensated AE and 1.52% for the temperature-compensated and temperature/humidity-compensated AEs. The reduction, 0.38 points, is a small difference — false alarms fell from 5 to 4 out of 263 evaluation samples — so its size is hard to emphasize. Still, on about nine months of full-scale bridge data with accumulated seasonal change and daily temperature ranges, the false-alarm rate moved down rather than up after compensation, so the direction of the temperature/humidity-compensation effect is consistent with our own model-bridge experiment. Adding humidity as a compensation variable did not lower the false-alarm rate below 1.52%, suggesting that the variation in openLAB's vibration characteristics over that period is explained mainly by temperature. The effect of humidity compensation was confirmed in our own model-bridge experiment, where the false-alarm rate in the humidity-change segment fell from 17.5% to 9.1%; that result is presented in 3.4.

This saturation of the improvement with temperature compensation alone is consistent with prior work [2], which reported that a bridge's natural frequency varies with temperature even without damage. A limitation is that openLAB's acceleration records are event-triggered, so the excitation source and amplitude differ from record to record and amplitude-dependent features vary greatly. Nevertheless, these differences in excitation act equally on all three models and the reconstruction-error procedure is identical, so the difference in false-alarm rate with and without compensation can be interpreted consistently.

3 Testing

3.1 Test Purpose

The tests have two purposes. First, to examine, with a self-built model bridge and low-cost sensors, how changes in mass distribution affect the vibration features and anomaly scores. The model bridge uses an ADXL355Z-based 3-axis MEMS accelerometer, a DHT22 temperature/humidity sensor and a Raspberry Pi 3 Model A+, with three mass conditions: 0-0, 1-0 and 1-1. This verifies whether low-cost sensors can collect and analyze vibration changes in structural conditions that differ from the normal baseline.

Second, to verify the effect of temperature/humidity compensation. The uncompensated, temperature-compensated and temperature/humidity-compensated models are applied with the same procedure to the openLAB dataset (see 2.7), public data from a full-scale bridge, to confirm first at real-bridge scale that environmental compensation reduces false alarms, and then to verify whether a similar trend of improvement is reproduced when the same compensation and detection pipeline is applied to data from our own model-bridge experiment. The openLAB analysis is thus not a separate plan replacing the model experiment but a reference comparison showing that the system's environmental-compensation method works consistently on both full-scale bridge data and our own experimental data; its process and results are presented in 2.7.

Design of the weight-based mass-distribution experiment

In the model-bridge experiment, real cracking or corrosion is not reproduced directly; instead, non-destructive changes in mass distribution are created with 10 g weights. Ten 10 g weights are stacked to make a 100 g load at each point, and up to twenty are used in the 1-1 condition, which uses both the left and right points. With the full length of the model bridge L , the left load point is set near $x = L/4$ and the right near $x = 3L/4$. Since the ADXL355Z accelerometer is attached at mid-span, $x = L/2$, the weights are placed so that they do not press on the sensor directly. There are three measurement conditions: 0-0, 1-0 and 1-1. 0-0 is the normal baseline with no weights, 1-0 adds 100 g to the left section, and 1-1 adds 100 g to each of the left and right sections. Because a single load on the right is structurally close to symmetric with the left condition, the single left load 1-0 is used as the representative condition. This experiment is interpreted not as a definitive reproduction of real cracking, corrosion or fatigue damage, but as a structural-condition test to see how the natural frequency, RMS, band energies and Autoencoder reconstruction error change with the mass distribution.

This analysis focuses on the data actually collected for the 100 g conditions 0-0, 1-0 and 1-1. If follow-up experiments judge that the change in peak frequency or anomaly score under the 100 g conditions is not large enough, an extended experiment raising the left and right loads to 200 g can be added. The extended experiment is not included in this analysis and is left as a future improvement.

Experiment type	Left section A	Right section B	Total combinations	Max weights needed
Basic experiment	0g / 100g	0g / 100g	4	20

Each experimental condition is recorded as left-right (0-0/1-0/1-1), where A is the left section and B the right section. In the basic experiment each digit is 0 = 0 g, 1 = 100 g, and the extended experiment additionally uses 2 = 200 g. For example, 0-0 is the normal baseline with no weights, and 1-0 is the state with 100 g on the left section only.

Condition label	Left A	Right B	Meaning
0-0	0g	0g	Normal baseline
1-0	100g	0g	Left mass increased
1-1	100g	100g	Left and right mass increased

Data acquisition and storage format

The data-acquisition program is configured to record x-, y- and z-axis acceleration from the ADXL355Z-based accelerometer at 200 Hz and temperature and relative humidity from the DHT22 at about 2-second intervals. Acceleration and temperature/humidity are saved in one file with the same elapsed-time value (ms) measured from the start of recording, so in analysis the vibration features extracted per time window can be linked by time to the mean temperature and mean relative humidity of that window.

Raw data are saved as CSV with columns for elapsed time (ms), DHT22 temperature and relative humidity, ADXL355Z internal temperature and 3-axis acceleration in g, and the left/right load condition and repetition number are recorded as per-file metadata. In analysis, per-axis vibration features (RMS, standard deviation, peak-to-peak, spectral entropy, peak frequency and power, band energies) are computed for each time window and linked to that window's mean temperature/humidity and load/environmental condition, and the peak frequency is used as the estimate of the first natural frequency.

3.2 Test Method

Testing is divided into analysis of the public openLAB data and our own model-bridge experiment. Because the entire measurement period of the openLAB data (see 2.7) is in the undamaged reference state, every threshold exceedance can be interpreted as a false alarm, allowing a direct comparison of the reduction in false alarms due to environmental compensation. The openLAB analysis compares the anomaly scores and threshold-exceedance rates of the uncompensated, temperature-compensated and temperature/humidity-compensated Autoencoders to confirm at real-bridge scale that environmental compensation reduces false alarms; the detailed process and results are presented in 2.7. Our own model-bridge experiment checks low-cost sensor data acquisition and the ability to detect changes in mass distribution, and verifies whether the compensation effect seen in the openLAB analysis is reproduced on our own data with the same pipeline.

The model-bridge experiment uses the ADXL355Z-based accelerometer and the DHT22 sensor. The ADXL355Z is attached at the mid-span position set in 2.2, and the DHT22 is installed beside the bridge in the same environment. The measurement targets are three mass-distribution conditions, 0-0, 1-0 and 1-1; in each condition, vibration is applied in the same way and acceleration and temperature/humidity are recorded for at least 20-30 seconds, repeated three or more times where possible. RMS, standard deviation, peak-to-peak, peak frequency, peak power, band energies and spectral entropy are extracted from the collected signals. Analysis was performed on a MacBook Pro (Apple M3 Max, 36 GB RAM, macOS 26) with Python 3.13, NumPy 2.3 and pandas 2.3, using acceleration sampled at 200 Hz, 4-second analysis windows (800 samples per window) and measurement CSV files for 9 conditions in total.

Initially, 12 conditions were planned, combining 3 mass conditions with 4 environmental conditions. However, the temperature-centered condition performed outdoors had a small temperature rise or was accompanied by humidity changes, so it could hardly be regarded as an independent temperature-change condition. The environmental conditions were therefore fixed at three — baseline environment (temperature and humidity not changed artificially), humidity-centered change, and simultaneous temperature/humidity change — and the results were organized around the 9 relatively reliable conditions combining these with the 3 mass conditions (0-0, 1-0, 1-1). The outdoor temperature-change data are used only as supplementary observations.

In analysis, the temperature/humidity compensation regression and the anomaly-detection model are trained only on data from the normal baseline condition 0-0, and the remaining conditions, including 1-0 and 1-1, are used only for evaluation. The anomaly threshold is set at the 99th percentile of the normal reconstruction error and applied identically to the uncompensated, temperature-compensated and temperature/humidity-compensated models, so that the compensation methods are compared fairly on the same basis.

3.3 Expected Results

In the 0-0 condition without weights, the normal baseline vibration pattern is expected to appear relatively stably. In 1-0, 100 g is added to the left section, so the mass distribution becomes asymmetric; in 1-1, a total of 200 g is added on both sides, so the effect of the overall mass increase is expected to be larger. Accordingly, the peak frequency, RMS, band

energies and Autoencoder reconstruction error may differ from the normal baseline. Physically, with constant stiffness, a natural frequency tends to fall as mass increases, so under the weight conditions a downward shift of the peak frequency from the normal baseline is expected first. In the 1-0 condition, where the mass distribution is asymmetric, the change in the shape of the vibration response may appear in amplitude or band-energy features before it appears as a frequency shift.

However, the range of temperature and humidity change that can be created indoors for the model bridge is smaller than the seasonal change experienced by openLAB, a full-scale outdoor bridge, over its nine-month monitoring period, so the compensation effect may be smaller than that seen in openLAB. The criterion for the model experiment is therefore not the absolute size of the compensation effect but whether the trend of improvement seen in openLAB (fewer false alarms after compensation) is reproduced in the same direction on our own system's data.

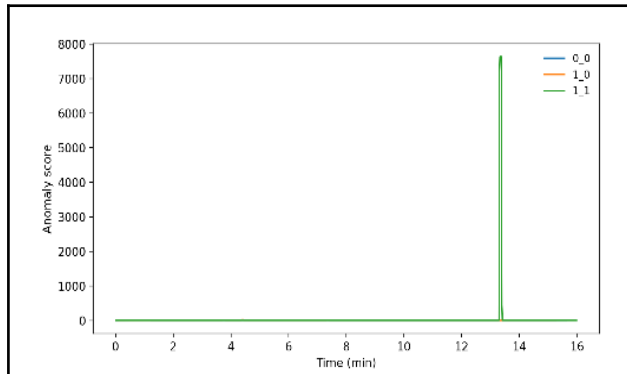
3.4 Actual Results

The actual results were compiled from the data collected for the 9 conditions. First, the mean peak frequency, rate of change in peak frequency, mean anomaly score, threshold-exceedance rate and main contributing band were compared for the 3 mass conditions (0-0, 1-0, 1-1); then the False Alarm Rate, Precision, Recall and F1-score of the uncompensated, temperature-compensated and temperature/humidity-compensated models were compared. The mean anomaly score and threshold-exceedance rate in the per-condition table aggregate all measurement windows of each condition (including all three environmental conditions) with the temperature/humidity-compensated model, while the False Alarm Rate in the model-comparison table counts only the normal windows with environmental variation in the evaluation period, so the rates in the two tables are not directly comparable. In the preliminary model test the first peak was observed at about 22.6 Hz, so to include it reliably the base bands of 2.3 (0-3, 3-10, 10-20 Hz) were reset to 0-12, 12-25 and 25-50 Hz. The main result plots — anomaly-score time series by condition, changes in the frequency spectrum, band-wise reconstruction-error contributions and so on — are presented in Figures 4-10. The +0.8% peak-frequency change in the 1-0 condition is opposite to the drop expected with added mass, but it is interpreted as a small change within the variation of manual excitation and the window-to-window fluctuation of the peak; the mass change of 1-0 was detected as an increase in the threshold-exceedance rate of the amplitude and band features (29.7%). In addition, looking only at the normal segment centered on humidity change, the false-alarm rate fell by nearly half, from 17.5% uncompensated to 9.1% with temperature/humidity compensation. This reduction (8.4 points) is contrary to the expectation in

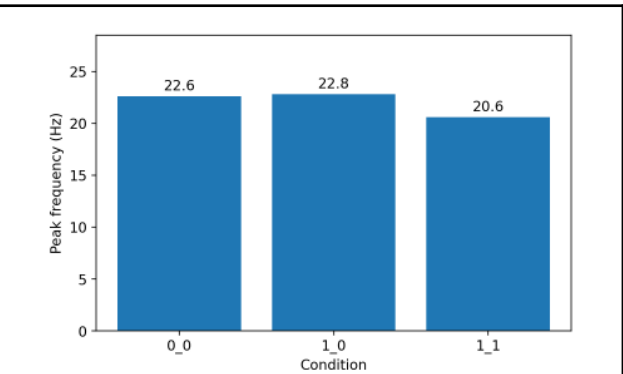
3.3 that it would be smaller than openLAB's (0.38 points); this is interpreted as follows: indoors, humidity could be changed artificially over a wide range, so the false-alarm rate before compensation was itself high and there was correspondingly more room for compensation to reduce it.

Condition ID	Left load	Right load	Mean f	Change in f	Mean anomaly score	Exceedance rate	Main contributing band
0-0	0g	0g	22.6 Hz	+0.0%	0.23	4.5%	0~12 Hz
1-0	100g	0g	22.8 Hz	+0.8%	0.37	29.7%	0~12 Hz
1-1	100g	100g	20.6 Hz	-8.9%	0.30	18.8%	12~25 Hz

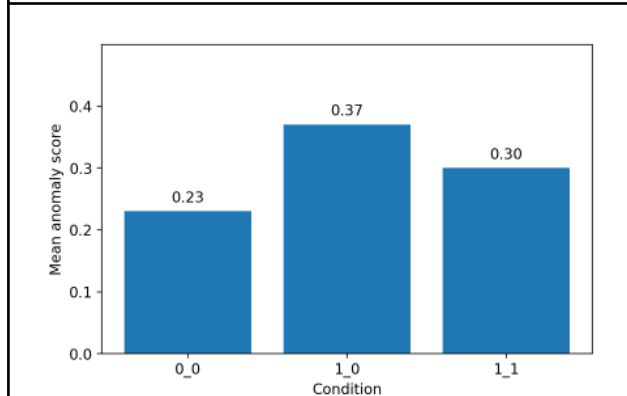
Model	Environmental inputs	False Alarm Rate	Precision	Recall	F1-score	Interpretation
Uncompensated AE	None	0.092	0.500	0.041	0.075	Vulnerable to environmental change
Temperature-compensated AE	Temperature T	0.048	0.892	0.197	0.323	Partly removes temperature effect
Temp/humidity-compensated AE	Temperature T + relative humidity RH	0.063	0.888	0.234	0.370	Compensates temp and humidity together



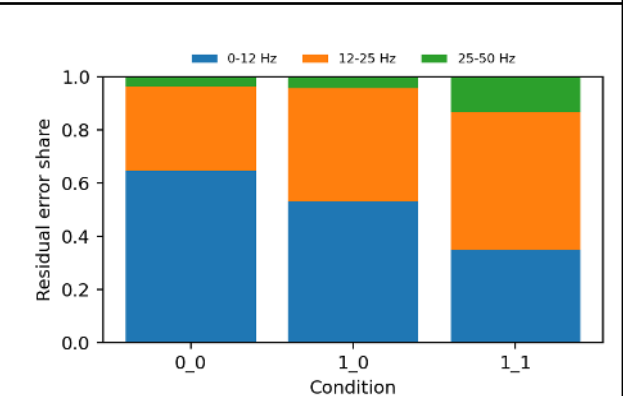
[Figure 4: Anomaly-score time series by condition]



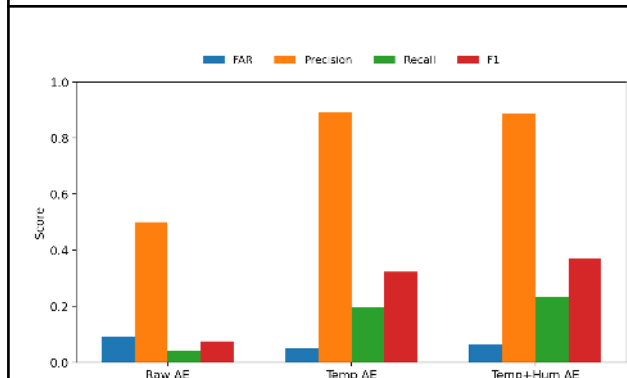
[Figure 5: Mean peak frequency by condition]



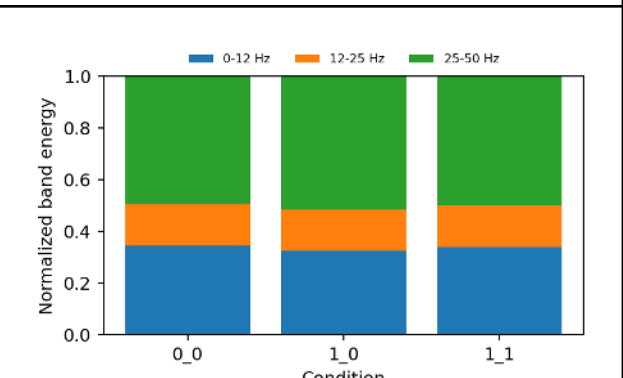
[Figure 6: Mean anomaly score by condition]



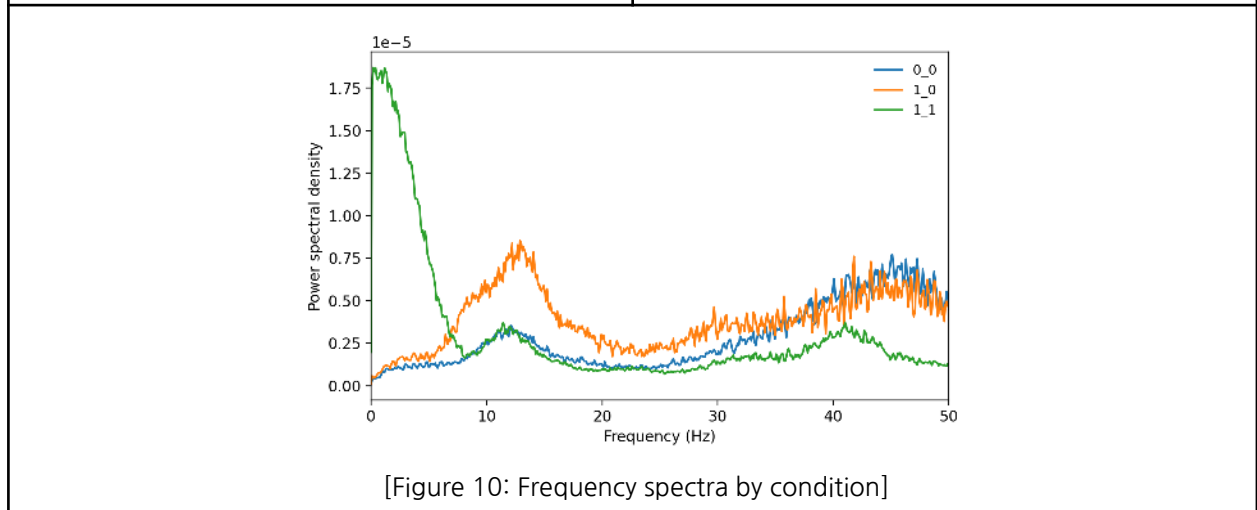
[Figure 7: Band reconstruction-error contribution by condition]



[Figure 8: Performance metrics by compensation step]



[Figure 9: Band-energy distribution by condition]



[Figure 10: Frequency spectra by condition]

3.5 Analysis of Results

The analysis first compared the peak frequency of the normal baseline 0-0 with each weight condition. By the rate of change in peak frequency, $\Delta f = (f_{\text{condition}} - f_{\text{baseline}}) / f_{\text{baseline}} \times 100 (\%)$, the 1-1 condition showed a clear drop of -8.9% , while 1-0 was detected not by a frequency change but by changes in RMS, the band-energy features and the Autoencoder reconstruction error. This shows that even a low-cost MEMS sensor can detect changes in dynamic properties due to changes in mass distribution. On a small foam-board model, however, differences in manual excitation, the sensor's fixing and cable movement may affect the natural frequency more than temperature and humidity do, so the temperature/humidity-compensation effect was not concluded from the model experiment alone but cross-checked against the openLAB analysis in 2.7; the result is reflected in the difference in exceedance rates between the uncompensated and temperature/humidity-compensated models.

In the band contributions, the main contributing band of the 1-1 condition, whose peak fell by 8.9% , was 12-25 Hz; this is the result of the first peak (about 22.6 Hz) changing the spectral shape of the band it belongs to, which is reflected in the reconstruction error, and is interpreted as the explanation agreeing with the physical change of a resonance shift. By contrast, the large contribution of the 0-12 Hz band in 0-0 and 1-0 is interpreted as relatively large variation in low-frequency energy and amplitude features. Band contributions can therefore give field inspectors a clue for distinguishing frequency-shift-type changes from amplitude-variation-type changes.

4 Interpretation & Conclusion

4.1 Conclusion

This work collected vibration and temperature/humidity data from a model bridge with a low-cost device consisting of an ADXL355Z accelerometer, a DHT22 sensor and a Raspberry Pi 3 Model A+, and checked whether the peak frequency, RMS, band energies and Autoencoder reconstruction error change under changes in mass distribution. After compensating for environmental effects with a temperature/humidity multivariate regression learned on normal data and judging anomalies by reconstruction error, the F1 score improved about 4.9-fold, from 0.075 uncompensated to 0.370, and the false-alarm rate in the normal segment where only humidity changed fell by nearly half, from 17.5% to 9.1%. Because manual excitation and the sensor's fixing also affect the model bridge, however, the temperature/humidity-compensation effect is interpreted cautiously, together with the openLAB analysis.

This improvement involves a trade-off. The false-alarm rate over all normal windows with environmental variation fell from 0.092 uncompensated to 0.048 with temperature compensation and then rose slightly to 0.063 with temperature/humidity compensation, while recall rose from 0.197 to 0.234, greatly improving F1, and precision on the damage conditions stayed high at about 0.89. By condition, in 1-1, with 100 g added on each side, the first peak frequency fell by 8.9% , so the large mass change clearly appeared as a resonance shift. In 1-0, with 100 g on one side only, the frequency change was only 0.8% , but the asymmetric mass made the response irregular, giving the highest anomalous-window rate, 29.7%, while 1-1 was actually lower at 18.8% as the first mode settled into a new resonance. Even in the normal condition 0-0 the anomalous-window rate remained at 4.5%, so false alarms did not disappear completely. In short, the system behaves as a first-stage screening tool: when it alarms, an actual anomaly is likely (high precision), but it catches only about 23% of all anomalies. Given these characteristics, the system is a supplementary means for aging small and medium bridges short of inspection staff and budget, using low-cost sensors to pick out suspect sections first and set precision-inspection priorities, and it serves as a selective early-warning means of anomaly risk rather than a replacement for precision safety diagnosis. Because the node is built from inexpensive general-purpose parts and is easy to distribute across many bridges, allocating inspection resources first to bridges where anomaly candidates and main contributing bands reported by each node accumulate can fill, at the level of continuous screening, the time gap between inspection cycles raised in 1.1.

What distinguishes this work from existing research lies in the form of the outcome. Its contribution is not only an analysis result obtained by applying an algorithm to public data, but the direct implementation of low-cost sensor-based data collection, environmental compensation, anomaly judgment and band-wise explanation as one measurement-and-analysis

pipeline. Vibration and temperature/humidity measurement, environmental compensation, anomaly judgment and band-contribution explanation were integrated into a single device that works at the measurement site, implemented and demonstrated directly; unlike fiber-optic BOCDA measurement, which precisely measures local deformation with dedicated infrastructure, it aims at low-cost screening applicable to small and medium aging bridges in general. It is also distinguished by explaining not just whether there is an anomaly but which frequency band departed from the normal baseline. Analysis of the long-term monitoring data of Germany's openLAB, public data from a full-scale bridge, was performed as a reference validation of this environmental-compensation method, and its results are presented in 2.7. It confirmed that environmental compensation reduces false alarms on real-bridge data with accumulated seasonal change and daily temperature ranges, and our own model-bridge experiment confirmed that the same pipeline reproduces a similar trend of improvement.

4.2 Future Improvements

This study identified three limitations. First, the foam-board model has high damping, so vibration dies out quickly and the first mode is unstable, making it hard to detect the small mass change of the 1-0 condition, with 100 g on one side only, as a frequency change. Second, the manual method of pressing and releasing with a finger applied vibration with inconsistent force and timing, increasing the variability of the data. Third, the range of temperature change that could be created indoors was small, limiting the effect of temperature-only compensation. Together these limitations kept damage-detection recall at about 0.23 even after compensation. The limitation of temperature range was supplemented by the openLAB analysis in 2.7, but that is offline validation, so follow-up research needs to extend the public-data analysis while improving our own system's excitation method, sensor layout and model structure to make real measurement data more stable. On this basis, three directions of improvement are planned. First, improve the model and experimental conditions. Replacing the foam board with aluminum beams removes the excessive damping and secures a stable first natural frequency, which can raise sensitivity and recall for small mass changes. Tapping automatically at a constant force and interval with a solenoid auto-tapper instead of a finger reduces input variation, narrows the feature distribution of normal data and makes the separation between normal and anomalous sharper. On this foundation, the experiment will be extended in three directions: finding the smallest detectable mass change with finer weight steps, examining the possibility of damage localization by dividing the model into sections, and verifying the compensation effect over a wider range of temperature and humidity. The goal of this stage is to show the halving of the humidity-segment false-alarm rate seen here (17.5% to 9.1%) in a clearer and more reproducible form.

Second, install the system on a real bridge in service for long-term field validation. A real bridge uses the ambient vibration produced by vehicles and wind as is, so no separate exciter or traffic closure is needed. Attaching measurement nodes to the deck or girders and collecting vibration and temperature/humidity continuously for weeks to months would let our own sensors directly observe the temperature-natural-frequency relationship that repeats daily and seasonally, quantify the reduction in false alarms from temperature/humidity compensation over a long period, and also test practical issues such as the durability of sensor mounting and the stability of power and storage.

Third, extend measurement beyond a single node to multiple nodes. With nodes at several points on a bridge, comparing responses between points makes it possible to examine the localization of anomalies, and collecting measurement data remotely could develop the system into a management tool that compares the anomaly scores of many bridges on one screen to help set precision-inspection priorities. In long-term operation, however, the baseline may drift because of sensor drift and similar effects, so a procedure for periodically retraining on normal data and updating the threshold is needed.

References

- [1] Caballero-Russi, D., Ortiz, A. R., Guzmán, A., & Canchila, C., "Design and Validation of a Low-Cost Structural Health Monitoring System for Dynamic Characterization of Structures," Applied Sciences, 12(6), 2807, 2022.
- [2] Peeters, B., & De Roeck, G., "One-year monitoring of the Z24-Bridge: environmental effects versus damage events," Earthquake Engineering & Structural Dynamics, 30(2), 149-171, 2001.
- [3] KU Leuven Structural Mechanics, "Z24 Bridge Benchmark," benchmark documentation, KU Leuven, Leuven, Belgium.
- [4] Korea Research Institute of Standards and Science (KRISS), "A 'smart sensor' that prevents bridge collapses and landslides," press release, 2019.11.12.
- [5] Giglioni, V. et al., "Deep Autoencoders for Unsupervised Damage Detection with Application to the Z24 Benchmark Bridge," IMAC, Springer, 2022.
- [6] Pakzad, S. S., & Masoodi, A. R., "Early damage detection in bridges using a variational autoencoder-based hybrid unsupervised learning framework," Scientific Reports, 16, 1389, 2026. DOI: 10.1038/s41598-025-31115-w
- [7] Aosong (Guangzhou) Electronics Co., Ltd., "DHT22/AM2302 Digital Temperature and Humidity Sensor," Datasheet.
- [8] Jansen, A., Richter, B., Röder, R., Herbers, M., & Marx, S., "Monitoring Data of the openLAB Research Bridge (2024-02-01 to 2024-10-31) and building information," OPARA - Open Access Repository and Archive, TU Dresden, 2025. DOI: 10.25532/OPARA-660.

Sources (list of materials)

No.	Type	Label in text	Description (keywords)	Notes (source, etc.)
1	Photo	Material 1	Jeongja Bridge walkway collapse site	https://www.anjunj.com/news/articleView.html?idxno=36589
2	Photo	Material 2	Fern Hollow Bridge collapse site	Taken by the NTSB (U.S. federal government work, public domain), posted on Wikimedia Commons
3	Figure	Material 3	Chart of the share of Class 3 bridges in Korea	Developed (from 2025 Korea Authority of Land & Infrastructure Safety statistics)
4	Figure	Material 4	Inspection cycles by facility grade	Developed (drawn from the inspection cycles under the Facility Safety Act)
5	Figure	Figures 1-3	Sensor layout, Autoencoder diagram, system flowchart	Developed (drawn by the authors)
6	Figure	Figures 4-10	7 plots of experimental results by condition and model	Developed (output of our own experimental data analysis)
7	Photo	2.6 Parts and model photos	ADXL355Z, DHT22, Raspberry Pi, foam-board model, finished module (5 photos)	Developed (taken by the authors)
8	Public data	openLAB	Long-term monitoring data from a full-scale bridge	TU Dresden OPARA (reference [8])
9	Open-source SW	-	Python, NumPy, pandas, Matplotlib (collection, analysis, visualization)	Official distributions (PSF/ BSD licenses)